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Cardinal Reserve: A Proposed Online Room Reservation System for a Higher Educational Institution (HEI)

System Analysis and Design of the Room Reservation of a Higher Educational Institution (HEI) Using Design Thinking Methodology

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ABSTRACT

Integrating information systems within organizations has proved to be significantly efficient, as it helps minimize the errors and delays caused by their present methods. In the field of education, automation is important to processes involving students such as room reservation. This study was conducted to analyze the present room reservation process of a higher education institution and come up with the design of an automated system that can decrease processing errors and provide online monitoring of request and feedback.

CCS CONCEPTS

• Information Systems; • Information Systems Applications;

KEYWORDS

Online reservation, Higher Education Institution, Information System

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1 INTRODUCTION

In the current dynamic educational field, the efficient management of physical resources within higher educational institutions

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stands as a foundation for fostering an optimal learning environment. Among these resources, classrooms or activity spaces serve as necessities where knowledge dissemination, collaboration, and innovation come together. For instance, Mapua University, known as a bastion of learning committed to providing an environment conducive to growth and development for its students, faculty, and staff. The room reservation system within higher educational institutions (HEI) plays a pivotal role in the utilization and allocation of these spaces. However, in an era characterized by rapid technological advancement and evolving educational patterns, the need to continuously assess and enhance such systems becomes evident.

Considering the volume of events, activities, and academic pursuits that happen within these institutions on a daily basis. Students, with their commitment to achieving excellence, arrange various engagements aimed at enriching the fabric of the educational community. Yet, the current manual process of reserving rooms and spaces for these activities serve crucial challenges. Students, organizations, and individuals navigating this traditional reservation process are faced with the necessary creation of paperwork, administrative issues, and uncertainty regarding the availability of these resources. The process from conceptualizing an event to securing a venue involves a complicated and lengthy process of documentation, approval, and verification, often leading to inefficiencies and delays.

With this context, this research seeks to make a transformative solution. Based on the principles of System Analysis and Design and supported by the concepts of Design Thinking, the idea to modernize and innovate the room reservation process in higher educational institutions came. The aim is not to just streamline administrative workflows but to change the way these institutions use technology to enhance the educational experience through the exploration of the challenges in the current reservation system and an understanding of the perspectives of end-users.

The modernization of the room reservation system aims to improve the educational experience, foster deeper student engagement and the potential of physical resources within these institutions. These efforts align with the broader institutional commitment to excellence and innovation, ensuring that higher education leads societal progress. In an era defined by rapid change and continuous innovation, the modernization of the room reservation system represents an operational and strategic enhancement.

2 REVIEW OF RELATED LITERATURE

2.1 Inefficiency in Current Reservation Process

Higher educational institutions' current room reservation process involves the use of the traditional and manual method which involves the use of paper. The current process of the room reservation is that the student has to personally attend the university to accomplish the reservation. The process is outdated and an inconvenience to both the student and admin due to the absence of the needed deans and administrators. This method additionally creates a huge problem for both sides because of the lack of ways to receive updates with the reservation. The lack of the digitization of the process impacts several factors, one which is a student's performance. The integration of digital technologies resulted into effective and efficient change [1]. However, it must be taken note that the adoption of digital technology relies on information flow and the adaptability of the user to adjust to the technology introduced by the institution [2]. A study in the relation of student leadership to digitization, the complexity of student leadership and digitization of processes through resources and development [3]. The need for digital competency and knowledge in being a student leader positively influences the implementation of digital technologies.

2.2 Lack of Digitized Process for Room Reservation

In improving reservation systems, a common thread among related studies is the digitization and automation of the process. A study utilized deep learning to predict classroom demand in higher education. Their deep learning algorithm analyzed historical data, enabling the accurate prediction of future room demand [4]. Similarly, another study introduced machine learning to enhance the efficiency of their school room reservation system. Machine learning processes data, creating algorithms and patterns that optimize room allocation for students, faculty, and administrators [5]. Evaluation in a simulated environment confirmed the approach's effectiveness in improving reservation system efficiency.

A smart school room reservation system incorporating the Internet of Things (IoT) and Artificial Intelligence (AI). This system featured IoT sensors to monitor room occupancy and an AI-based reservation system providing real-time reservation information [6]. The addition of a mobile application allowed students and teachers to view room availability and make reservations. Real-world deployment of this design effectively increased room utilization and reduced nonattendance to reservations.

To secure systems handling diverse data such as student numbers, Verma employed blockchain technology, ensuring data integrity and tamper prevention. a chatbot-based reservation system for school rooms, offering a user-friendly experience. This chatbot interacted with students and users using natural language, successfully implemented and evaluated at a local high school. These innovative technologies enhance the efficiency, security, and user experience of reservation systems in educational settings [6].

3 METHODS

3.1 Current Service Blueprint

"Understanding customer experience and the customer journey over time is critical," [7] Customer journey maps were created to gain a more comprehensive understanding of the customer experience. Its primary purpose is to capture and visualize touchpoints and interactions a customer had the entire customer journey. In connection to this, service blueprints are fundamental in fulfilling the satisfaction goal for the customer experience. This tool helps designers visualize the end-to-end service process, including the frontstage and backstage activities, customer interactions and touchpoints. This blueprint helps map out the customer journey, enabling designers to identify pain points, areas for improvement and opportunities for innovation. Moreover, this analysis has led to the elimination of tasks, reduction of movements, and overall simplification of work [8]. The researchers used the tools to create an automated system that will achieve its goal of providing a more convenient and easier platform for its users to reserve a room.

Figure 1 shows the current service blueprint of the room reservation system. The process starts when the student council or organization members discuss their event if either online or face-to-face. After choosing their modality, they must search the campus for rooms and to check its availability. They now need to discuss the reservation and the room that they are choosing from. They will then finalize the details for their activity and chosen room. A student council or organization representative will be getting a copy of the room reservation form which can be taken from the Admin Office. The representative will fill out the form with the necessary and correct information with their fellow student council or organization officers. They will then proceed to the Faculty Room to ask for the approval and signature of their adviser. Once they get their adviser's signature, the representative will proceed to the Admin Office to pass the registration form. The administrator tells the representative to come back after several days to know the result of their room reservation. In circumstances where the administrator or the adviser is not present, the representative needs to visit another day for approval, passing, or confirmation.

Trying to reserve a room within the campus for various student-organized activities, the chosen institution uses a manual system to ensure that these rooms will be accounted for. The following are the pain points of the current system: the use of paper forms to make official reservations which causes delays in the process of implementing student activities which lessens the productivity of the students. Manual reservation makes the travel of information limited as it only circles between a single group of people, unless thoroughly asked by other parties, resulting in an inadequate amount of information when it comes to the knowledge of availability of the location they have in mind for their event.

4 DATA COLLECTION

The researchers conducted a survey and made use of a list of questions to curate crucial data regarding the chosen higher educational institution's current room reservation system. The method of disseminating the questionnaire was through Google Forms, which is currently one of the most widely used platforms for surveys and gathering important online data. The researchers designed a

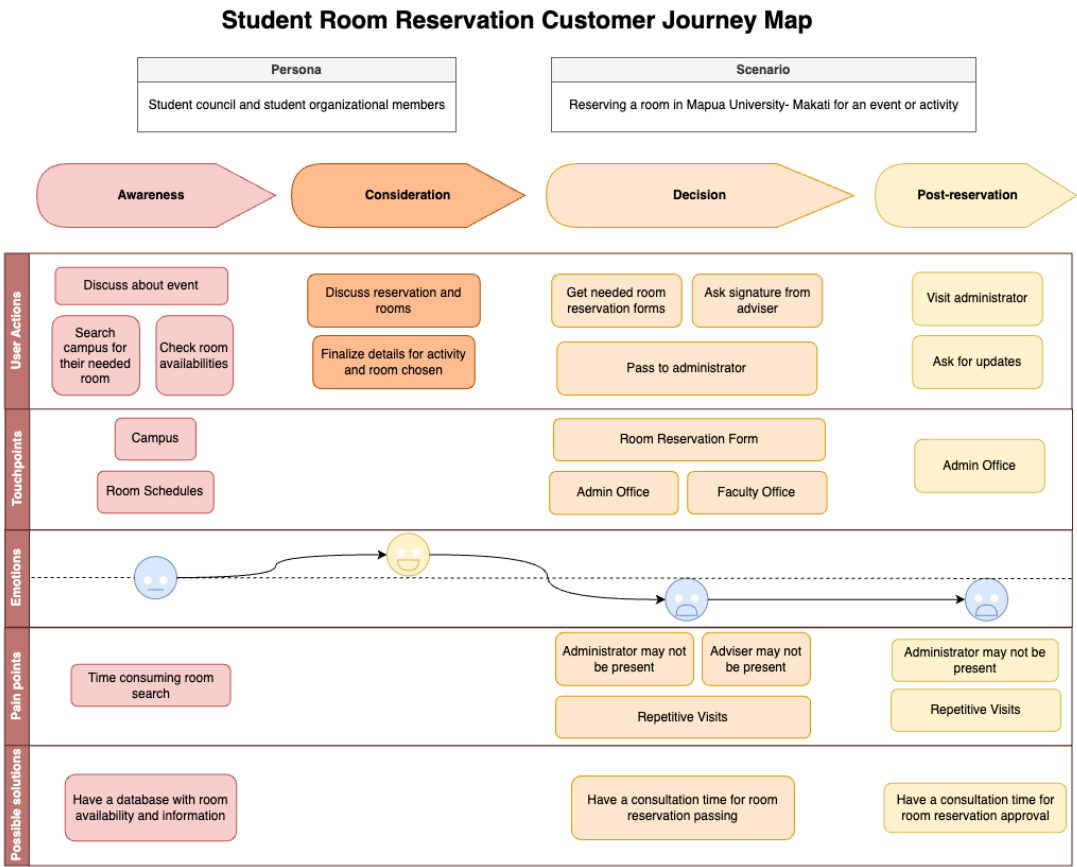


Figure 1: Customer Journey Map of Online Room Reservation

questionnaire consisting of various types of questions, including multiple-choice, scale questions, and open-ended questions. The respondents for the survey would be student council members and organization members, as they would have first-hand experience of reserving a facility for their organization’s objectives and plans which resulted in 50 respondents in the survey, providing a diverse range of perspectives.

The questionnaire addressed the perception of the students about the impacts of the current room reservation system towards the school community. According to the results, the students frequently reserve rooms to fulfill their organization’s plans or events. As a result, they need to file a reservation request for the room at least 1-2 weeks in advance, to ensure that they can confirm its availability. It would take some time and effort for the students to receive information about their request, which would cause delays. The students found it difficult to modify their requests, but concerns were still addressed to be solved.

The scale question part of the survey addressed the impact of the current room reservation process towards its users. Most respondents found the current system satisfactory enough to meet their specific needs and criterion in reserving a desired facility. However, they are also aware that an automated approach to the system is

still a necessity, given that it can help improve the current processes and increase the speed of their reservation request.

Finally, the open-ended questions part of the survey focused on a personalized experience of the users with regards to the room reservation process of the institution. The respondents’ purpose of securing a facility is for organizational-related activities and events. They acknowledged the fact that the current process itself is inefficient and time-consuming, given that manual reservation is laborious and causes delays. The central issues experienced by the students in the implementation of the current processes were the following: difficulty of obtaining signatures, confirmations and approvals, change in room availability, processing of papers, absence of the necessary people for approval, late addressing of concerns, and reservation conflicts or delays. The respondents recognize the need for automation for the improvement of the efficiency in the process, and minimize the risks and errors indicated. The creation of a user-friendly system will increase both productivity and engagement within the community for the primary focus on bettering the experience of both the students and administrators regarding room reservation.

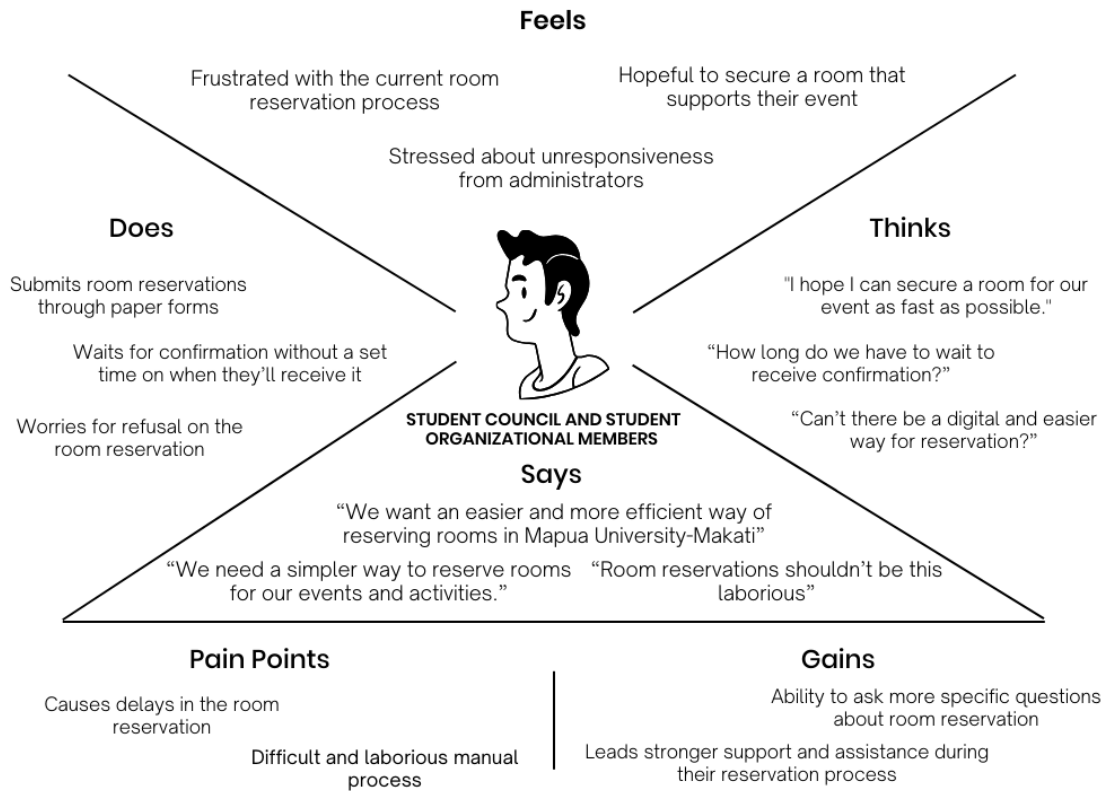


Figure 2: Empathy Map

4.1 Results and Discussions

Mapua University-Makati presently employs a manual process for room reservations, which has posed obstacles in institutional events. This issue not only impacts students' perceptions regarding the room reservation system but also creates an inconvenience during the reservation process. A positive perception could encourage individuals to utilize the manual process, while negative perceptions might deter them from satisfaction.

The following information shows how users feel about the program, their thoughts and experiences, and how it affects their overall happiness with Manual Room Reservation. The empathy map is used to learn about the user's sentiments, actions, motives, and concerns about the software. Along with the "Point of View" and "How Might We," more context will be supplied in developing answers to problems encountered by student council members and student organization members in the current room reservation system.

4.1.1 Empathy Maps. Figure 2 shows the frustrations and difficulties that student council and organizational members experience when obtaining rooms at the higher educational institution. Students face delays, confusion, and stress as a result of the present approach, which depends on paper forms and lacks clear communication schedules.

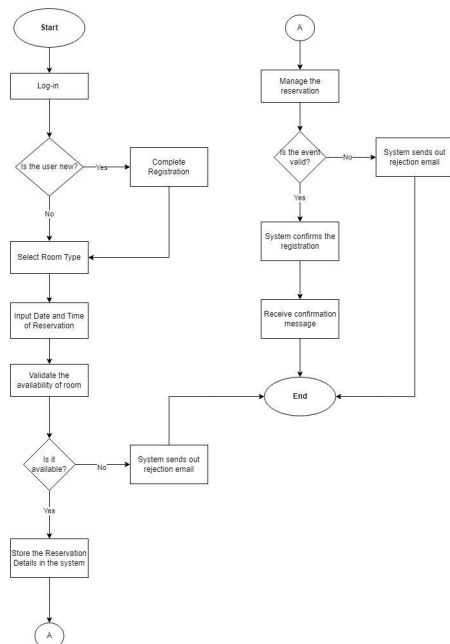
Students are particularly frustrated by the time-consuming manual process and officials' lack of reaction. They are optimistic about finding a room that can accommodate their gathering, but they are also concerned about potential refusals. Their opinions show a desire for a speedier, more efficient process, and their queries reveal a need for better communication and timeframes.

The empathy map also reveals that students want a simpler and more efficient approach to reserve rooms, and they are willing to acquire the help they want in room reservation. Overall, the empathy map gives a great glimpse of student council and student organizational members' experiences with room reservation at higher educational institutions. This data may be utilized to create and implement solutions that enhance the room reservation process for students, such as a digital room reservation system or more responsive administration communication.

The Empathy Map serves as a tool for constructing business models from the standpoint of clients' perspectives. Its template offers a visually structured layout, facilitating its implementation. In addition, the Empathy Map comprises guiding questions that help designers create personas in a methodical manner [9]. The use of empathy maps revealed that the successful utilization of it enabled the formulation of designs, regardless of their differing levels of empathy [10].

Table 1: POVs and HMWs for Each Persona

Insights	Needs	POV	HMW
Users find it difficult to immediately determine whether the schedule they have aligns with the room they wish to reserve.	Users need a lesser time-consuming way to figure out if the schedule they have aligns with the room they wish to reserve.	The student council member must be able to determine the availability of the room so that their planning for the activity will go smoothly.	How might we let the student council member immediately know whether the room is available or not?
Users are unable to always be physically present at the institution to reserve a room	Users need a more convenient way to reserve a room, without compromising their personal agendas by going to school.	The student organization member needs to be able to reserve a room without compromising his/her personal agendas by going to school.	How might we help the student organization member to conveniently reserve a room anytime, anywhere?

**Figure 3: Proposed System Flowchart for Online Room Reservation**

4.1.2 POVs and HMWs for Each Persona. As seen in Table 1 below, there are two personas that the survey itself has curated whom are specifically, student council and organization members:

4.1.3 Proposed System Flowchart. Figure 3 shows the proposed system for the room reservation which aims to automate and improve on the current process of reservation, providing convenience and efficiency for the community of the chosen institution. The user will log in to the online system. If they are new to the system, they are prompted to complete their registration. If they're not,

then they are directed to selecting their preferred room type. Afterwards, they will input the date and time of their desired reservation. Through the calendar provided by the system, the room availability will be validated based on the given schedule. If the room is not available, a message regarding its unavailability will be sent to notify the user. On the other hand, if the room is confirmed to be available, the given reservation details are stored within the system. These details would include the room number/location, date and time of reservation, and the equipment needed. Moving forward, the system administrators are able to access and monitor the reservation dashboard, which allows them to prepare accordingly. If the system administrators decide that the reason behind the event is not as important, then a message will be sent to the user relaying the reservation's rejection. However, if the system admins find the reason behind the event's requirement for reservation valid, then they will officially confirm its availability, generating a message with a unique ID. Finally, the student receives the confirmation via their given email, or through a reminder from the system.

4.1.4 Use Case Diagram. Figure 4 shows the interaction of both the students and administrators with the system to accomplish the room reservation itself. The student uses the system to first log into the system, then views the room schedule and checks its details, before providing his/her reservation proposal. Afterwards, they will reserve the room, manage the room details (indicate their required equipment, etc.). They have the option to cancel the room reservation should they feel the need to. Students will also receive updates regarding their reservation, whether it is confirmed or not. They will then log-out after performing the aforementioned tasks. On the other hand, the administrators are also inclined to log into the system, and either approve or decline the reservations, by browsing the pending requests. They will then give the reservation updates to the students through the system, and will also log-out after doing so.

4.1.5 Hierarchy plus Input-Process-Output (HIPO) Chart. Figure 5 shows the processes involved in the online room reservation process at a high level. First, the user enters important event information such as the date, time, and event type. Following that, the system performs a room availability validation, confirming whether or not the requested room is available for reservation. The system delivers a confirmation message to the user after confirming availability. The

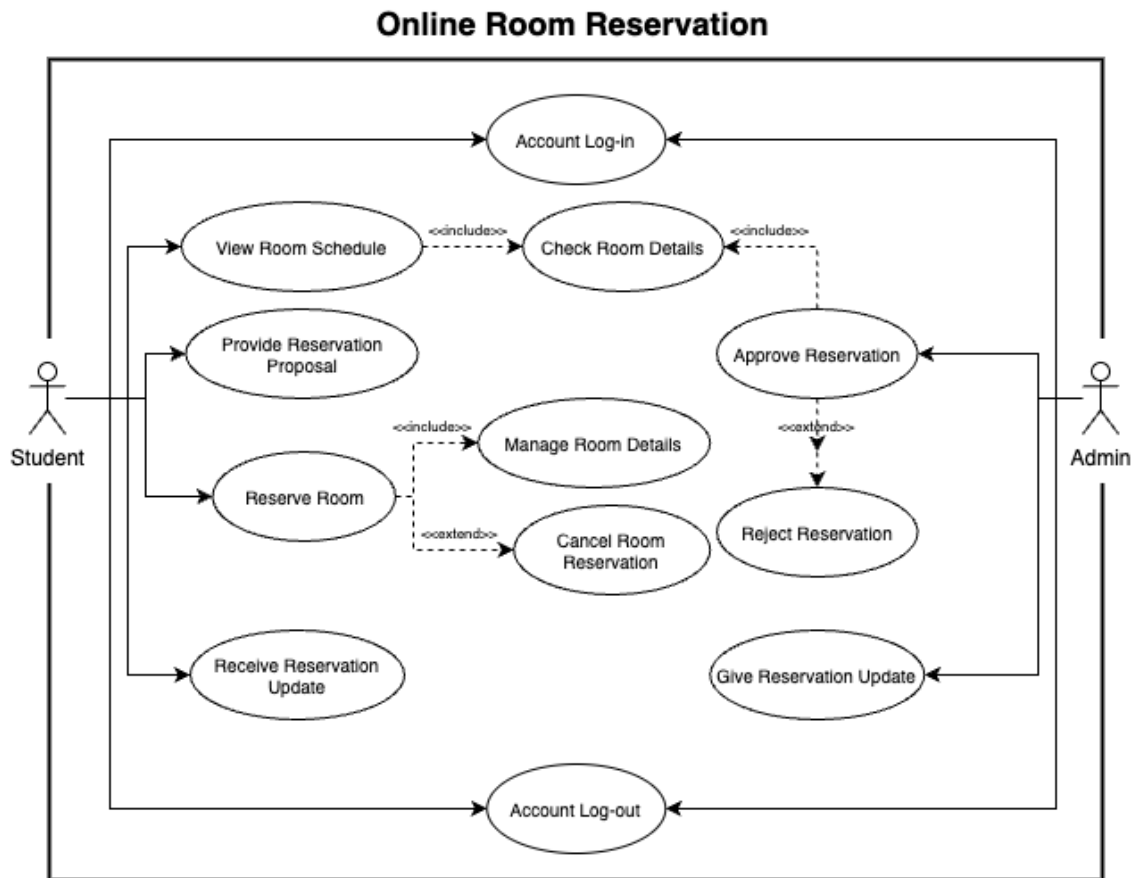


Figure 4: Use Case Diagram

user then picks their preferred room and the number of participants for the event.

The system then does a schedule validation to ensure that the room is available at the desired time and date, and if successful, generates another confirmation for the user. Simultaneously, the system allows the user to pick the appropriate equipment for their event. An inventory check is performed to assess the availability of the equipment, and if the equipment is available, a confirmation message is delivered to the user. This series of stages efficiently walks consumers through the process of making a lodging reservation.

Event Details: The first step in making an online room reservation is to enter event information. This provides details such as the event's name, date, hour, and projected attendance. The event information is used to assess room and equipment availability.

Room Availability and Capacity: Following the submission of the event information, the system will verify the availability of rooms that satisfy the provided parameters. This involves double-checking the room's date, hour, and capacity. If a suitable room is available, the system will hold it for a set length of time to allow the user to finish the reservation.

Equipment Availability: If a user wants an item of equipment, the system will verify its availability. This involves double-checking the equipment's date, time, and amount. If the equipment is available, the system will hold it for a set amount of time to allow the user to finish the reservation.

4.1.6 Data Flow Diagram for Online Room Reservation. Figure 6 shows the data flowing between the user, system, and administrator which are as follows: account details, list of rooms, room type details, schedule details, reservation details, confirmation details, and the confirmation message itself. The process starts with a login prompt of the user, where they are inclined to input their account details, beginning the data flow between the user and the system. Afterwards, the system shows the user the available room types within the establishment. Users are then able to choose which type of room suits their event's needs and specifications. They will input their choice of room, as well as their desired date and time. These details will then be stored and compiled within the system, which makes way for the data to move on from the system to the administrators. The admins will receive the reservation details from the system itself, allowing them to review the specifics of everything then will either confirm or reject the reservation

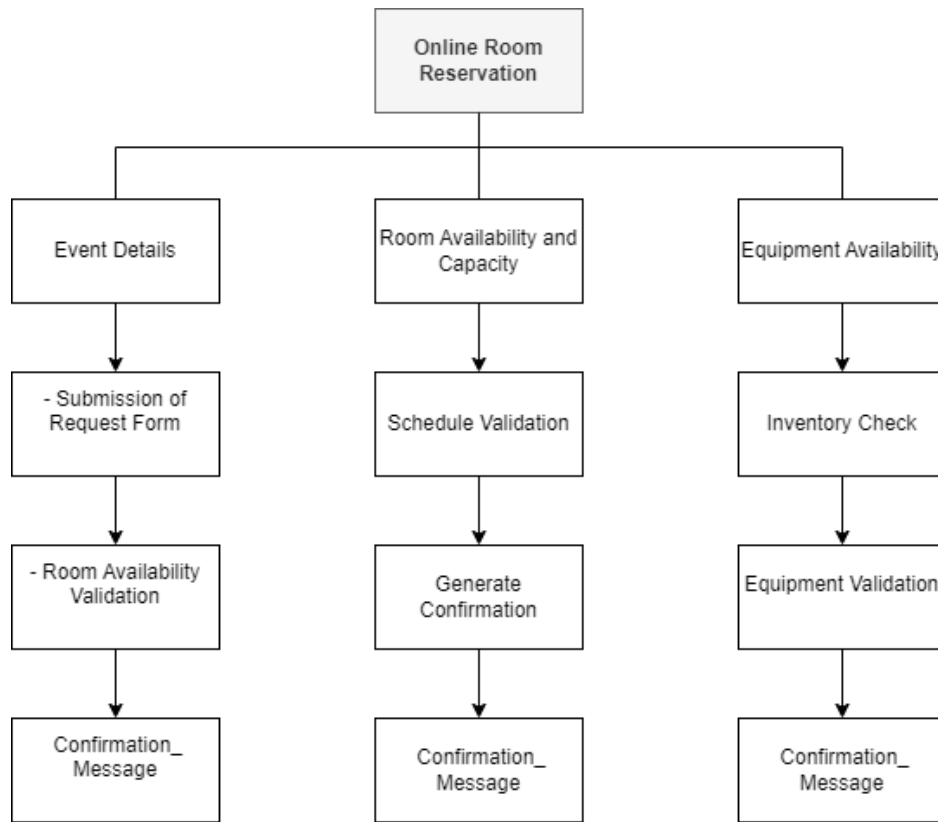


Figure 5: Hierarchy plus Input-Process-Output (HIPO) Chart

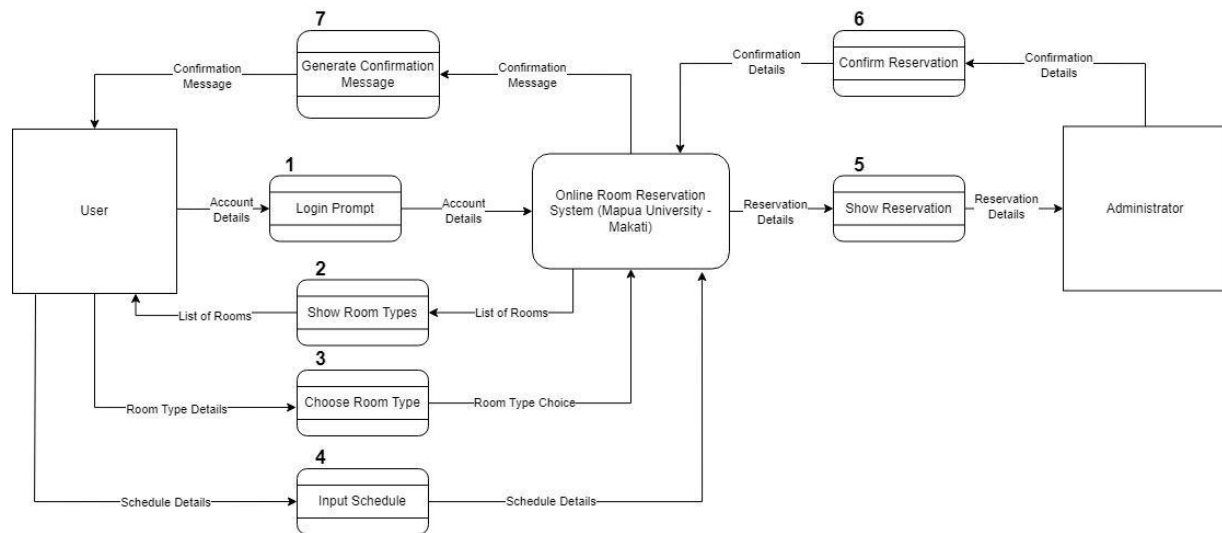


Figure 6: Data Flow Diagram for Online Room Reservation

request, but in this case, the reservation was confirmed. Once done, the administrators will produce the confirmation details and send it through the system, which will then initiate the network to

generate the confirmation message, sending it back to the user for their information.

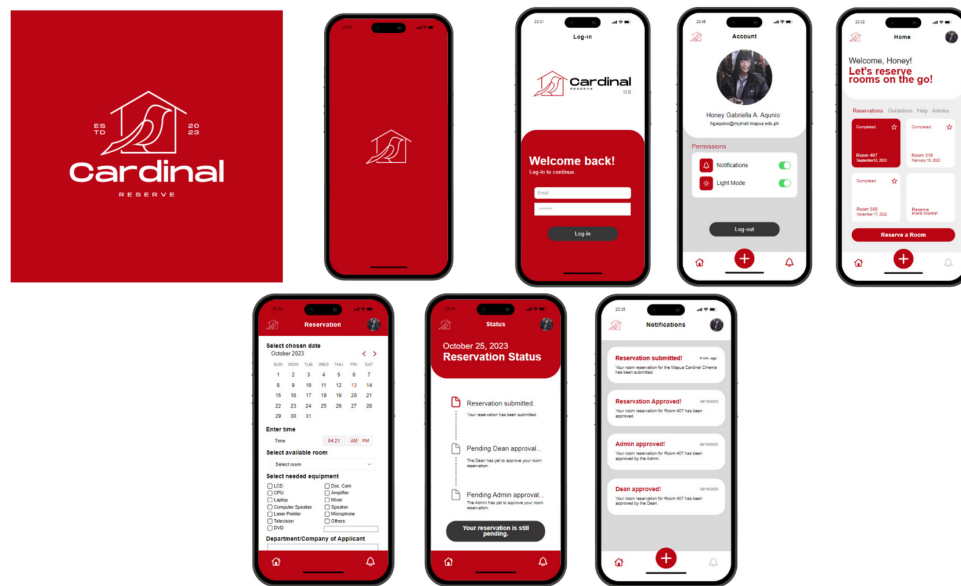


Figure 7: User Interface

4.1.7 User Interface. Figure 7 shows the user interface of the student when accessing the proposed mobile application for the proposed online room reservation system. When accessing the application, the user is asked to log-in with their institutional email which would be linked to the institution's system. After the log-in page, the user would be greeted by the home page which includes access to past and current reservations, notifications, and a button to reserve a room. When the button is clicked, the user will be directed to the reservation page where the questions found in the manual room reservation paper form is reflected in an online form. If the user submits a reservation request, their ongoing reservation will show in the home page and when clicked will show the status if it has been approved by the institution's administration. The user will also receive notifications about approval and if their reservation has been completed.

Figure 8 shows the user interface of the administration side of the proposed online room reservation system. The user will also be prompted to log-in using their institutional email then directed to the home page. The home page shows the user their dashboard which includes several categories, such as "Needs Approval," "Rejected," and "Approved" where the user has the option to approve more rooms, check the details of already approved and rejected rooms. The "Needs Approval" tile when clicked shows the user room reservations that require their approval. They have the option to check the information of the reservation and a button to quickly approve the request. The user would also be receiving notifications regarding new room reservation requests that need their approval.

4.1.8 System Evaluation. Table 2 shows the tabularized comparison of both current and proposed processes for the room reservation system. Currently, the higher educational institution makes use of paper forms to properly document and officiate the reservation of

rooms for certain events. Students are asked to fill up a form that would specify which room they are to reserve, for what purpose it is to be used, when it will be used, and which equipment is needed to fulfill the said event. The signature of the student council or organization member would be needed, along with the approval of their adviser and the department's dean and/or program chair. Finally, when all information is completed, they would simply pass the paper to the admin office and wait for the approval of the admins before executing the next phase of their event planning. Usually, the confirmation or rejection would take as early as hours or as late as weeks, which is inconvenient for the students and cause delays in the implementation of certain activities which decreases productivity. Furthermore, the manual reservation of rooms leaves the information only to a single group of people. Should there be another organization that plans on implementing an event on the exact same date, it would take time for them to know that another organization already reserved the room or location that day, leaving room for misinformation and disorientation within the school community.

The researchers' proposed online system would be the automation of the said system with the aim to improve on the current processes implemented within the institution and provide convenience and increased productivity. The proposed online room reservation system would include the use of an application that allows its users, particularly the student council and organization members, admins, to reserve a room at any given time and place. The user logs in to the system and is given the option to choose the room with the date and time of their event. After this, the information will be sent through the system, where the admin will also be able to access it and determine the approval of the event.

The proposed online room reservation offers numerous advantages. First, it will increase the productivity of everyone within

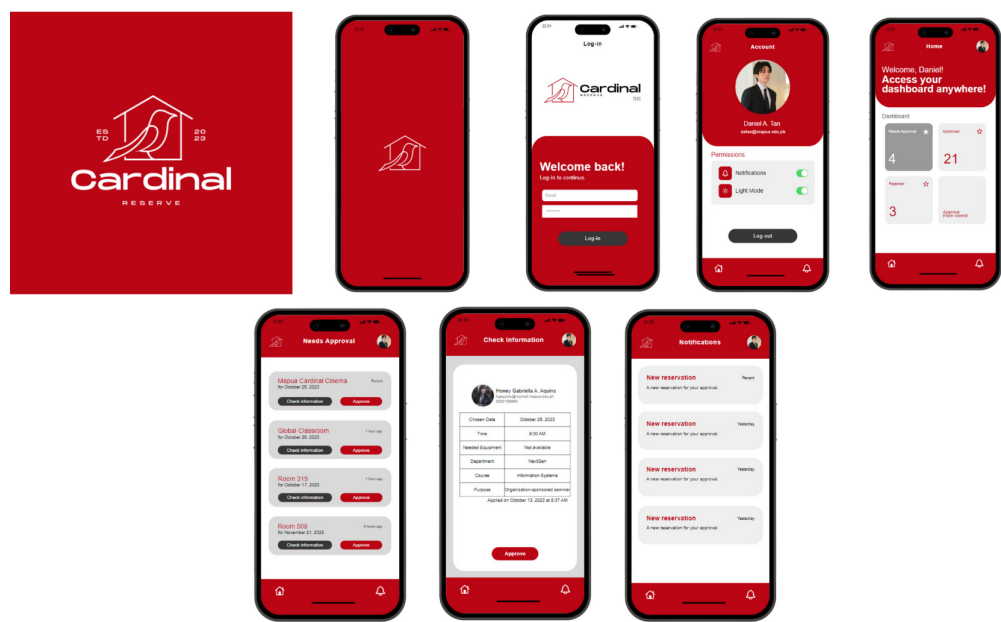


Figure 8: Admin Interface

Table 2: System Evaluation for Cardinal Reserve

System Aspect	Current Process	Proposed Process	Time Comparison
User Interface	Makes use of paper forms and physical submission	Makes use of online application, which is accessible via mobile and web	Process is faster and more convenient
Reservation Details	Crucial information such as room, purpose, date and equipment is handwritten	Online input of room type, date and time selection through a form	Faster and more secure data entry
Approval Workflow	Manual submission, multiple signatures required	Admin access, feasibility evaluation through the app	Time for data flow is significantly reduced
Decision Time	Approval takes hours to weeks	Faster decision-making of up to at least minutes to an hour, real-time updates in the app	Reduced approval time
Impact on Productivity	Causes delays in event planning, reducing student productivity	More convenient room reservation, increasing productivity	Improved productivity, reduced delays
Information Sharing	Limited awareness of existing reservations	Real-time room availability, reduced risk of conflicts	Enhanced awareness

it in a long-term setting and make the process easier. It will also enhance the user experience, where it will become more convenient for them to reserve a room. Additionally, the automated system reduces the need for manual data entry and processing. Through this, workload is lessened, and information would be easily stored and documented. And finally, 24/7 availability and real-time updates, users can conveniently access up-to-the-minute room availability information, reducing the risk of overbookings and misinformation.

5 CONCLUSION

To conclude, the proposed online room reservation system presents a significant technological advancement in the institution. It leverages the available resources to optimize the reservation process, making it more efficient, convenient, and accessible for all students, faculty, and administrators. By automating the process, it reduces the need for manual paperwork and speeds up the approval process, significantly improving productivity and reducing delays in event implementation.

The system offers real-time room availability information, instant notifications, and the ability to make reservations from anywhere, at any time, simplifying the process for all parties involved. Moreover, the system provides a digital trail of reservation information, making it easier to track and manage room bookings. Overall, the system aligns with the modernization efforts of educational institutions and enhances the overall experiences of the people in the institutions, making room reservations more user-friendly and efficient.

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A APPENDICES

A SURVEY QUESTIONNAIRE

As part of our research, we conducted a survey to gather relevant information from the users regarding the feasibility of the current system. Here, we included a list of various questions (see Appendix A.1) used in the study, along with the data privacy consent request (see Appendix A.2) that participants were requested to confirm.

For the survey results, see 4. **DATA COLLECTION segment** above.

A.1 - LIST OF QUESTIONS

A.1.1 Multiple Choice Questions.

- How frequently do you reserve a room at Mapua Makati?
- How far in advance do you typically make room reservations?
- Are there specific hours or days when you prefer to make room reservations?
- How long does it take for you to fully confirm that the room you reserved is available?
- Are there any parts of the room reservation process that were difficult?
- Are you able to easily find information about room availability?
- Do you receive timely information about your room reservation?
- Are you able to modify or cancel room reservations easily?
- Did you seek help or had concerns during the room reservation process? If yes, were your concerns addressed?

A.1.2 Scale Questions. **Scale: 1 - Strongly Disagree | 2 - Disagree | 3 - Neutral | 4 - Agree | 5 - Strongly Agree**

- The room reservation procedure at Mapua Makati is user-friendly and accessible.
- The present room reservation procedure at Mapua Makati is highly satisfactory.
- The communication regarding modifications or revisions to accommodation reservations is efficient and quick.
- The confirmation of my room reservation was sent to me promptly.
- The employee or the staff assigned to room reservation is helpful and responsive during the reservation process.

- The availability of the rooms met expectations and needs.
- The manual process of room reservation is easily understandable.
- The room reservation system was efficient and timely.
- Special requests or preferences (e.g., accessibility restrictions, unique accommodation characteristics) were fulfilled well by the reservation system.

A.1.3 Open-ended Questions.

- What is usually your main purpose for reserving a room within the school?
- Can you tell us how the present manual reservation method affects your productivity or efficiency?
- What are some of the issues or difficulties you have experienced while manually reserving rooms at Mapua Makati?
- Do you have any suggestions or recommendations for the improvement of the room reservation?

A.2 - DATA PRIVACY ACT OF 2012 (CONSENT REQUEST)

Room Reservation in Mapua University-Makati

Mapua University advocates for student activities that help raise engagement within the community, ensuring a productive type of atmosphere for everyone. As such, students are inclined to reserve rooms for their intended events. In the process of room reservation, the researchers aim to find areas of improvement within the system itself, in order to give way to better opportunities and for easier reservation of rooms.

The purpose of this survey is to evaluate the current room reservation system, identify any challenges or areas for improvement, and gather suggestions from respondents on how we can enhance the overall experience of reserving rooms on campus.

Rest assured that your responses will be kept confidential and will only be used for the purpose of this research.

Data Privacy

This questionnaire is following the Data Privacy Act of the Philippines (Republic Act No. 10173). The data gathered will be used for educational purposes only, sensitive information such as name and email will not be collected.

Data Privacy Act of 2012 Any member of the organization and its accredited partners may share these data between them and such data may constitute my personal or sensitive personal information (which includes without limitation, my name, email, age, education information, and other information classifies as "personal data", "personal information", or "sensitive personal information" under the Data Privacy Act.)

This online survey is approximately 5 to 10 minutes. Your basic information, name, email address, or IP will not be collected for your anonymity. All of the information that is acquired is for research purposes only, and they will be held with confidentiality.

Acknowledgement *

☐ I agree and acknowledge about the purpose of this questionnaire and the data I will share.